Linear Approximation and L'Hôpital's Rule

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Tangent Line Approximation (Linearization)**

Use linearization (the tangent line) of $f(x) = 7\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{10})$.

Final Answer: $\frac{287\sqrt{2}}{40}$. $L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = 7\sqrt{2}$, $f'(2) = \frac{7\sqrt{2}}{4}$, so $L(2 + \frac{1}{10}) \approx \frac{287\sqrt{2}}{40}$.

2 **[Foundational]** • **Target: L'Hôpital's Rule for 0/0 Forms**

Use linearization (the tangent line) of $f(x) = 6\sqrt{x}$ at $x = 2$ to approximate $f(2 - \frac{1}{10})$.

Final Answer: $\frac{117\sqrt{2}}{20}$. $L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = 6\sqrt{2}$, $f'(2) = \frac{3\sqrt{2}}{2}$, so $L(2 - \frac{1}{10}) \approx \frac{117\sqrt{2}}{20}$.

3 **[Foundational]** • **Target: L'Hôpital's Rule for Infinity/Infinity Forms**

Use linearization (the tangent line) of $f(x) = 6\sqrt{x}$ at $x = 5$ to approximate $f(5 - \frac{1}{10})$.

Final Answer: $\frac{297\sqrt{5}}{50}$. $L(x) = f(5) + f'(5)(x - 5)$. Here $f(5) = 6\sqrt{5}$, $f'(5) = \frac{3\sqrt{5}}{5}$, so $L(5 - \frac{1}{10}) \approx \frac{297\sqrt{5}}{50}$.

4 **[Foundational]** • **Target: Comparing Relative Growth Rates of Functions**

Use linearization (the tangent line) of $f(x) = 7\sqrt{x}$ at $x = 5$ to approximate $f(5 - \frac{1}{5})$.

Final Answer: $\frac{343\sqrt{5}}{50}$. $L(x) = f(5) + f'(5)(x - 5)$. Here $f(5) = 7\sqrt{5}$, $f'(5) = \frac{7\sqrt{5}}{10}$, so $L(5 - \frac{1}{5}) \approx \frac{343\sqrt{5}}{50}$.

5 **[Foundational]** • **Target: Tangent Line Approximation (Linearization)**

Use linearization (the tangent line) of $f(x) = 7\sqrt{x}$ at $x = 2$ to approximate $f(2 - \frac{1}{10})$.

Final Answer: $\frac{273\sqrt{2}}{40}$. $L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = 7\sqrt{2}$, $f'(2) = \frac{7\sqrt{2}}{4}$, so $L(2 - \frac{1}{10}) \approx \frac{273\sqrt{2}}{40}$.

6 **[Foundational]** • **Target: L'Hôpital's Rule for 0/0 Forms**

Use linearization (the tangent line) of $f(x) = 3\sqrt{x}$ at $x = 5$ to approximate $f(5 + \frac{1}{5})$.

Final Answer: $\frac{153\sqrt{5}}{50}$. $L(x) = f(5) + f'(5)(x - 5)$. Here $f(5) = 3\sqrt{5}$, $f'(5) = \frac{3\sqrt{5}}{10}$, so $L(5 + \frac{1}{5}) \approx \frac{153\sqrt{5}}{50}$.

7 **[Foundational]** • **Target: L'Hôpital's Rule for Infinity/Infinity Forms**

Use linearization (the tangent line) of $f(x) = 6\sqrt{x}$ at $x = 5$ to approximate $f(5 + \frac{1}{5})$.

Final Answer: $\frac{153\sqrt{5}}{25}$. $L(x) = f(5) + f'(5)(x - 5)$. Here $f(5) = 6\sqrt{5}$, $f'(5) = \frac{3\sqrt{5}}{5}$, so $L(5 + \frac{1}{5}) \approx \frac{153\sqrt{5}}{25}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Comparing Relative Growth Rates of Functions

Use linearization (the tangent line) of $f(x) = 5\sqrt{x}$ at $x = 3$ to approximate $f(3 + \frac{1}{10})$.

Final Answer: $\frac{61\sqrt{3}}{12}$. $L(x) = f(3) + f'(3)(x - 3)$. Here $f(3) = 5\sqrt{3}$, $f'(3) = \frac{5\sqrt{3}}{6}$, so $L(3 + \frac{1}{10}) \approx \frac{61\sqrt{3}}{12}$.

9 [Intermediate] • Target: Tangent Line Approximation (Linearization)

Use linearization (the tangent line) of $f(x) = 7\sqrt{x}$ at $x = 3$ to approximate $f(3 - \frac{1}{5})$.

Final Answer: $\frac{203\sqrt{3}}{30}$. $L(x) = f(3) + f'(3)(x - 3)$. Here $f(3) = 7\sqrt{3}$, $f'(3) = \frac{7\sqrt{3}}{6}$, so $L(3 - \frac{1}{5}) \approx \frac{203\sqrt{3}}{30}$.

10 [Intermediate] • Target: L'Hôpital's Rule for 0/0 Forms

Use linearization (the tangent line) of $f(x) = 5\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{10})$.

Final Answer: $\frac{41\sqrt{2}}{8}$. $L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = 5\sqrt{2}$, $f'(2) = \frac{5\sqrt{2}}{4}$, so $L(2 + \frac{1}{10}) \approx \frac{41\sqrt{2}}{8}$.

11 [Intermediate] • Target: L'Hôpital's Rule for Infinity/Infinity Forms

Use linearization (the tangent line) of $f(x) = 4\sqrt{x}$ at $x = 3$ to approximate $f(3 + \frac{1}{5})$.

Final Answer: $\frac{62\sqrt{3}}{15}$. $L(x) = f(3) + f'(3)(x - 3)$. Here $f(3) = 4\sqrt{3}$, $f'(3) = \frac{2\sqrt{3}}{3}$, so $L(3 + \frac{1}{5}) \approx \frac{62\sqrt{3}}{15}$.

12 [Intermediate] • Target: Comparing Relative Growth Rates of Functions

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{10})$.

Final Answer: $\frac{41\sqrt{2}}{20}$. $L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = 2\sqrt{2}$, $f'(2) = \frac{\sqrt{2}}{2}$, so $L(2 + \frac{1}{10}) \approx \frac{41\sqrt{2}}{20}$.

13 [Intermediate] • Target: Tangent Line Approximation (Linearization)

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 3$ to approximate $f(3 - \frac{1}{10})$.

Final Answer: $\frac{59\sqrt{3}}{30}$. $L(x) = f(3) + f'(3)(x - 3)$. Here $f(3) = 2\sqrt{3}$, $f'(3) = \frac{\sqrt{3}}{3}$, so $L(3 - \frac{1}{10}) \approx \frac{59\sqrt{3}}{30}$.

14 [Intermediate] • Target: L'Hôpital's Rule for 0/0 Forms

Use linearization (the tangent line) of $f(x) = 5\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{5})$.

Final Answer: $\frac{21\sqrt{2}}{4}$. $L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = 5\sqrt{2}$, $f'(2) = \frac{5\sqrt{2}}{4}$, so $L(2 + \frac{1}{5}) \approx \frac{21\sqrt{2}}{4}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: L'Hôpital's Rule for Infinity/Infinity Forms • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 2$ to approximate $f(2 - \frac{1}{10})$.

Final Answer: $\frac{39\sqrt{2}}{20}$. $L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = 2\sqrt{2}$, $f'(2) = \frac{\sqrt{2}}{2}$, so $L(2 - \frac{1}{10}) \approx \frac{39\sqrt{2}}{20}$.

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16 [Challenging] • Target: Comparing Relative Growth Rates of Functions • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 5$ to approximate $f(5 - \frac{1}{10})$.

Final Answer: $\frac{99\sqrt{5}}{50}$. $L(x) = f(5) + f'(5)(x - 5)$. Here $f(5) = 2\sqrt{5}$, $f'(5) = \frac{\sqrt{5}}{5}$, so $L(5 - \frac{1}{10}) \approx \frac{99\sqrt{5}}{50}$.

17 [Challenging] • Target: Tangent Line Approximation (Linearization) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Use linearization (the tangent line) of $f(x) = 5\sqrt{x}$ at $x = 5$ to approximate $f(5 - \frac{1}{10})$.

Final Answer: $\frac{99\sqrt{5}}{20}$. $L(x) = f(5) + f'(5)(x - 5)$. Here $f(5) = 5\sqrt{5}$, $f'(5) = \frac{\sqrt{5}}{2}$, so $L(5 - \frac{1}{10}) \approx \frac{99\sqrt{5}}{20}$.

18 [Challenging] • Target: L'Hôpital's Rule for 0/0 Forms • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Use linearization (the tangent line) of $f(x) = 3\sqrt{x}$ at $x = 3$ to approximate $f(3 + \frac{1}{5})$.

Final Answer: $\frac{31\sqrt{3}}{10}$. $L(x) = f(3) + f'(3)(x - 3)$. Here $f(3) = 3\sqrt{3}$, $f'(3) = \frac{\sqrt{3}}{2}$, so $L(3 + \frac{1}{5}) \approx \frac{31\sqrt{3}}{10}$.

19 [Challenging] • Target: L'Hôpital's Rule for Infinity/Infinity Forms • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 5$ to approximate $f(5 - \frac{1}{5})$.

Final Answer: $\frac{49\sqrt{5}}{25}$. $L(x) = f(5) + f'(5)(x - 5)$. Here $f(5) = 2\sqrt{5}$, $f'(5) = \frac{\sqrt{5}}{5}$, so $L(5 - \frac{1}{5}) \approx \frac{49\sqrt{5}}{25}$.

20 [Challenging] • Target: Comparing Relative Growth Rates of Functions • Cross-Lesson: 8.01 Area Between Curves

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{5})$.

Final Answer: $\frac{21\sqrt{2}}{10}$. $L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = 2\sqrt{2}$, $f'(2) = \frac{\sqrt{2}}{2}$, so $L(2 + \frac{1}{5}) \approx \frac{21\sqrt{2}}{10}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Tangent Line Approximation (Linearization)

What is the local linear approximation $L(x)$ of $f(x) = \sqrt{x}$ at $x = 25$?

a. $L(x) = 5 + \frac{1}{10}(x - 25)$ ✓

b. $L(x) = 5 + \frac{1}{5}(x - 25)$

c. $L(x) = 5 + \frac{1}{2}(x - 25)$

d. $L(x) = \frac{1}{10}x$

$f(25) = 5$. $f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow f'(25) = \frac{1}{10}$. Linearization: $L(x) = f(a) + f'(a)(x - a) = 5 + \frac{1}{10}(x - 25)$.

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22 [Foundational] • Target: Tangent Line Approximation (Linearization)

Using the linearization of $f(x) = \sqrt{x}$ at $a = 25$, approximate $\sqrt{26}$.

a. 5.10 ✓

b. 5.05

c. 5.20

d. 5.01

$L(26) = 5 + 0.1(26 - 25) = 5 + 0.1 = 5.10.$

23 [Foundational] • Target: Tangent Line Approximation (Linearization)

Is the tangent line approximation for $f(x) = \sqrt{x}$ an overestimate or underestimate, and why?

a. **An overestimate because $f''(x) < 0$ (concave down)** ✓

b. An underestimate because $f''(x) > 0$ (concave up)

c. Exact because f is monotonic

d. Depends on the step size

Since $f''(x) = -\frac{1}{4}x^{-3/2} < 0$, the curve is concave down and lies below its tangent lines, making tangent line approximations overestimates.

24 [Foundational] • Target: L'Hôpital's Rule for 0/0 Forms

Evaluate $\lim_{x \to 0} \frac{e^{2x} - 1}{\sin(3x)}$.

a. $\frac{2}{3}$ ✓

b. $\frac{3}{2}$

c. 0

d. ∞

Direct substitution yields $\frac{0}{0}$. By L'Hôpital's Rule: $\lim_{x \to 0} \frac{2e^{2x}}{3\cos(3x)} = \frac{2(1)}{3(1)} = \frac{2}{3}$.

25 [Foundational] • Target: L'Hôpital's Rule for 0/0 Forms

Evaluate $\lim_{x \to 1} \frac{\ln x}{x^2 - 1}$.

a. $\frac{1}{2}$ ✓

b. 1

c. 2

d. 0

$\frac{0}{0}$ form. By L'Hôpital's: $\lim_{x \to 1} \frac{1/x}{2x} = \frac{1}{2(1)} = \frac{1}{2}$.

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26 [Foundational] • Target: L'Hôpital's Rule for 0/0 Forms

Evaluate $\lim_{x\to 0} \frac{1-\cos(4x)}{x^2}$.

a. 8 ✓

b. 4

c. 2

d. 16

First L'Hôpital's: $\frac{4\sin(4x)}{2x}$. Second L'Hôpital's: $\frac{16\cos(4x)}{2} = \frac{16(1)}{2} = 8$.

27 [Foundational] • Target: L'Hôpital's Rule for Infinity/Infinity Forms

Evaluate $\lim_{x\to\infty} \frac{\ln(x)}{x}$.

a. 0 ✓

b. 1

c. ∞

d. Undefined

$\frac{\infty}{\infty}$ form. By L'Hôpital's: $\lim_{x\to\infty} \frac{1/x}{1} = 0$.

28 [Foundational] • Target: L'Hôpital's Rule for Infinity/Infinity Forms

Evaluate $\lim_{x\to\infty} \frac{x^2}{e^x}$.

a. 0 ✓

b. 1

c. 2

d. ∞

Apply L'Hôpital's twice: $\lim_{x\to\infty} \frac{2x}{e^x} = \lim_{x\to\infty} \frac{2}{e^x} = 0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: L'Hôpital's Rule for Infinity/Infinity Forms

Which requirement MUST be verified before applying L'Hôpital's Rule to $\lim_{x\to c} \frac{f(x)}{g(x)}$?

a. **The limit produces an indeterminate form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$** ✓

b. $f(c) = 0$ only

c. $f'(c) = 0$

d. $g(x) > 0$ for all x

L'Hôpital's Rule applies ONLY when the quotient evaluates to an indeterminate form of type $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$.

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30 [Intermediate] • Target: Comparing Relative Growth Rates of Functions

As $x \to \infty$, which of the following functions grows the fastest?

a. e^x ✓

b. x^{10}

c. $x \ln x$

d. 2^x

Exponentials with larger base grow faster than any polynomial or power function. Since $e \approx 2.718 > 2$, e^x grows fastest.

31 [Intermediate] • Target: Comparing Relative Growth Rates of Functions

Evaluate $\lim_{x \to \infty} \frac{(\ln x)^5}{\sqrt{x}}$.

a. 0 ✓

b. ∞

c. 1

d. 5

Any power of x (even $x^{1/2}$) grows strictly faster than any power of $\ln x$. Hence the limit is 0.

32 [Intermediate] • Target: Comparing Relative Growth Rates of Functions

If $\lim_{x \to \infty} \frac{f(x)}{g(x)} = \infty$, what does this conclude about their growth rates?

a. $f(x)$ **grows faster than** $g(x)$ ✓

b. $g(x)$ grows faster than $f(x)$

c. They grow at the same rate

d. Neither function grows

By definition of relative growth rates, $\lim_{x \to \infty} \frac{f(x)}{g(x)} = \infty$ indicates that $f(x)$ grows strictly faster than $g(x)$.

33 [Intermediate] • Target: Tangent Line Approximation (Linearization)

Use linearization (the tangent line) of $f(x) = 1\sqrt{x}$ at $x = 2$ to approximate $f(2 - \frac{1}{10})$.

a. $-1 + \frac{39\sqrt{2}}{40}$

b. $\sqrt{2}$

c. $1 + \frac{39\sqrt{2}}{40}$

d. $\frac{39\sqrt{2}}{40}$ ✓

$L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = \sqrt{2}$, $f'(2) = \frac{\sqrt{2}}{4}$, so $L(2 - \frac{1}{10}) \approx \frac{39\sqrt{2}}{40}$.

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34 [Intermediate] • Target: L'Hôpital's Rule for 0/0 Forms

Use linearization (the tangent line) of $f(x) = 1\sqrt{x}$ at $x = 3$ to approximate $f(3 + \frac{1}{10})$.

a. $\sqrt{3}$

b. $\frac{61\sqrt{3}}{60}$ ✓

c. $1 + \frac{61\sqrt{3}}{60}$

d. $-1 + \frac{61\sqrt{3}}{60}$

$L(x) = f(3) + f'(3)(x - 3)$. Here $f(3) = \sqrt{3}$, $f'(3) = \frac{\sqrt{3}}{6}$, so $L(3 + \frac{1}{10}) \approx \frac{61\sqrt{3}}{60}$.

35 [Intermediate] • Target: L'Hôpital's Rule for Infinity/Infinity Forms

Use linearization (the tangent line) of $f(x) = 1\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{10})$.

a. $\sqrt{2}$

b. $-1 + \frac{41\sqrt{2}}{40}$

c. $\frac{41\sqrt{2}}{40}$ ✓

d. $1 + \frac{41\sqrt{2}}{40}$

$L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = \sqrt{2}$, $f'(2) = \frac{\sqrt{2}}{4}$, so $L(2 + \frac{1}{10}) \approx \frac{41\sqrt{2}}{40}$.

36 [Intermediate] • Target: Comparing Relative Growth Rates of Functions

Use linearization (the tangent line) of $f(x) = 1\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{5})$.

a. $1 + \frac{21\sqrt{2}}{20}$

b. $\frac{21\sqrt{2}}{20}$ ✓

c. $\sqrt{2}$

d. $-1 + \frac{21\sqrt{2}}{20}$

$L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = \sqrt{2}$, $f'(2) = \frac{\sqrt{2}}{4}$, so $L(2 + \frac{1}{5}) \approx \frac{21\sqrt{2}}{20}$.

37 [Intermediate] • Target: Tangent Line Approximation (Linearization)

Use linearization (the tangent line) of $f(x) = 6\sqrt{x}$ at $x = 3$ to approximate $f(3 + \frac{1}{10})$.

a. $-1 + \frac{61\sqrt{3}}{10}$

b. $\frac{61\sqrt{3}}{10}$ ✓

c. $6\sqrt{3}$

d. $1 + \frac{61\sqrt{3}}{10}$

$L(x) = f(3) + f'(3)(x - 3)$. Here $f(3) = 6\sqrt{3}$, $f'(3) = \sqrt{3}$, so $L(3 + \frac{1}{10}) \approx \frac{61\sqrt{3}}{10}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: L'Hôpital's Rule for 0/0 Forms • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Use linearization (the tangent line) of $f(x) = 4\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{10})$.

a. $\frac{41\sqrt{2}}{10}$ ✓

b. $1 + \frac{41\sqrt{2}}{10}$

c. $4\sqrt{2}$

d. $-1 + \frac{41\sqrt{2}}{10}$

$L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = 4\sqrt{2}$, $f'(2) = \sqrt{2}$, so $L(2 + \frac{1}{10}) \approx \frac{41\sqrt{2}}{10}$.

39 [Challenging] • Target: L'Hôpital's Rule for Infinity/Infinity Forms • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Use linearization (the tangent line) of $f(x) = 4\sqrt{x}$ at $x = 2$ to approximate $f(2 - \frac{1}{10})$.

a. $\frac{39\sqrt{2}}{10}$ ✓

b. $-1 + \frac{39\sqrt{2}}{10}$

c. $1 + \frac{39\sqrt{2}}{10}$

d. $4\sqrt{2}$

$L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = 4\sqrt{2}$, $f'(2) = \sqrt{2}$, so $L(2 - \frac{1}{10}) \approx \frac{39\sqrt{2}}{10}$.

40 [Challenging] • Target: Comparing Relative Growth Rates of Functions • Cross-Lesson: 8.01 Area Between Curves

Use linearization (the tangent line) of $f(x) = 4\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{5})$.

a. $1 + \frac{21\sqrt{2}}{5}$

b. $-1 + \frac{21\sqrt{2}}{5}$

c. $4\sqrt{2}$

d. $\frac{21\sqrt{2}}{5}$ ✓

$L(x) = f(2) + f'(2)(x - 2)$. Here $f(2) = 4\sqrt{2}$, $f'(2) = \sqrt{2}$, so $L(2 + \frac{1}{5}) \approx \frac{21\sqrt{2}}{5}$.

41 [Challenging] • Target: Tangent Line Approximation (Linearization) • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Use linearization (the tangent line) of $f(x) = 6\sqrt{x}$ at $x = 4$ to approximate $f(4 + \frac{1}{10})$.

a. $\frac{263}{20}$

b. $\frac{223}{20}$

c. $\frac{243}{20}$ ✓

d. 12

$L(x) = f(4) + f'(4)(x - 4)$. Here $f(4) = 12$, $f'(4) = \frac{3}{2}$, so $L(4 + \frac{1}{10}) \approx \frac{243}{20}$.

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42 [Challenging] • Target: L'Hôpital's Rule for 0/0 Forms • Cross-Lesson: 1.01 Limits Graphically and Numerically

Use linearization (the tangent line) of $f(x) = 6\sqrt{x}$ at $x = 4$ to approximate $f(4 - \frac{1}{5})$.

a. $\frac{107}{10}$

b. 12

c. $\frac{117}{10}$ ✓

d. $\frac{127}{10}$

$L(x) = f(4) + f'(4)(x - 4)$. Here $f(4) = 12$, $f'(4) = \frac{3}{2}$, so $L(4 - \frac{1}{5}) \approx \frac{117}{10}$.

43 [Challenging] • Target: L'Hôpital's Rule for Infinity/Infinity Forms • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Use linearization (the tangent line) of $f(x) = 7\sqrt{x}$ at $x = 1$ to approximate $f(1 - \frac{1}{10})$.

a. $\frac{153}{20}$

b. $\frac{133}{20}$ ✓

c. 7

d. $\frac{113}{20}$

$L(x) = f(1) + f'(1)(x - 1)$. Here $f(1) = 7$, $f'(1) = \frac{7}{2}$, so $L(1 - \frac{1}{10}) \approx \frac{133}{20}$.

44 [Challenging] • Target: Comparing Relative Growth Rates of Functions • Cross-Lesson: 2.01 The Definition of the Derivative

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 4$ to approximate $f(4 - \frac{1}{10})$.

a. $\frac{59}{20}$

b. $\frac{99}{20}$

c. $\frac{79}{20}$ ✓

d. 4

$L(x) = f(4) + f'(4)(x - 4)$. Here $f(4) = 4$, $f'(4) = \frac{1}{2}$, so $L(4 - \frac{1}{10}) \approx \frac{79}{20}$.

45 [Challenging] • Target: Tangent Line Approximation (Linearization) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Use linearization (the tangent line) of $f(x) = 1\sqrt{x}$ at $x = 4$ to approximate $f(4 - \frac{1}{10})$.

a. $\frac{119}{40}$

b. $\frac{39}{40}$

c. 2

d. $\frac{79}{40}$ ✓

$L(x) = f(4) + f'(4)(x - 4)$. Here $f(4) = 2$, $f'(4) = \frac{1}{4}$, so $L(4 - \frac{1}{10}) \approx \frac{79}{40}$.

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